

# Unit 2 REVIEW

1 The weight will travel three times as far during the second second as during the first.

2 a Acceleration =  $\frac{\text{resultant force}}{\text{total mass}}$

$$\therefore a = \frac{(20 - 10) \times (9.80)}{30} = 3.3 \text{ ms}^{-2} \text{ clockwise}$$

i.e.  $3.3 \text{ ms}^{-2}$  up for the 10 kg mass and  $3.3 \text{ ms}^{-2}$  down for the 20 kg mass.

b The resultant force on the  $F_a = F_g + T$

$$ma = mg + T$$

$$20 \times (-3.3) = 20 \times (-9.80) + T$$

$$T = -66 + 196$$

$$T = 1.3 \times 10^2 \text{ N}$$

3 The steady force applied by the engine is equal and opposite to the combined resistance forces such as air resistance and friction between the wheels and track. The net resultant force on the carriages is zero, and according to Newton's first and second laws, constant velocity is the result.

4 a  $v^2 = u^2 + 2as$

$$(10.0)^2 = (5.00)^2 + 2a \times (100)$$

$$\therefore a = 0.375 \text{ ms}^{-2} \text{ west}$$

b  $\Sigma F = ma = 1000 \times 0.375 \text{ west} = 375 \text{ N west}$

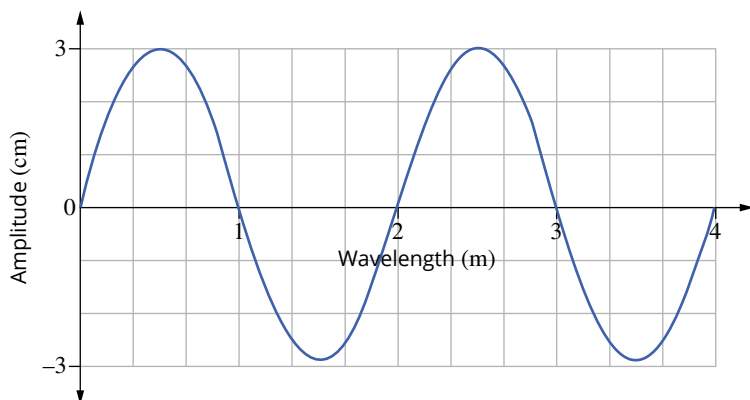
c  $\Sigma F =$  force exerted by tow truck – frictional force

$$375 \text{ N} = \text{force exerted by tow truck} - 200 \text{ N}$$

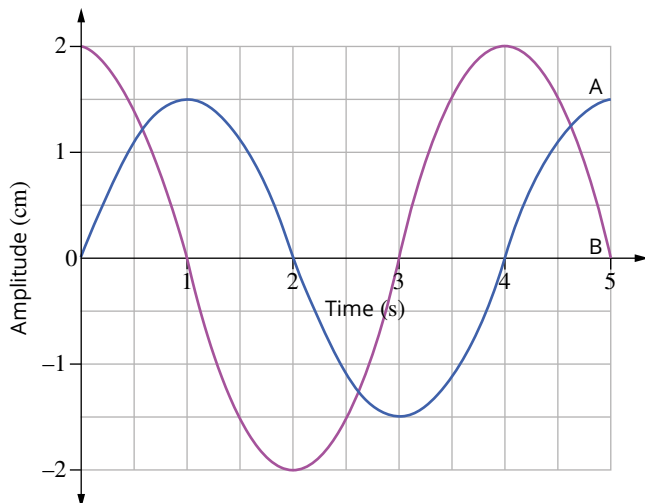
$$\therefore \text{force exerted by tow truck} = 375 \text{ N} + 200 \text{ N} = 575 \text{ N west}$$

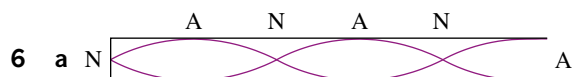
d The tow truck and car form an action–reaction pair, so the car exerts the opposite force: 575 N east.

5 a



b





b  $\lambda = \frac{4\ell}{(2n-1)} = \frac{4 \times 1.14}{1} = 4.56 \text{ m}$

$f = \frac{v}{\lambda} = \frac{346}{4.56} = 75.9 \text{ Hz}$

- 7 The line from Jenny to the speaker in front of her and the line between the speakers make a right angle. Applying Pythagoras' theorem, the distance from Jenny to the other speaker is 3.90 m. The path distance between Jenny and the two speakers is therefore 0.30 m. For constructive interference, the path difference between Jenny and the two speakers must be an odd multiple of half wavelengths.

Therefore  $\frac{\lambda}{2} = 0.30$ , so  $\lambda = 0.60 \text{ m}$

or  $\frac{3\lambda}{2} = 0.30$ , so  $\lambda = 0.20 \text{ m}$

or  $\frac{5\lambda}{2} = 0.30$ , so  $\lambda = 0.12 \text{ m}$

and  $f = \frac{v}{\lambda}$ , therefore for the first three frequencies,  $f = 576.7 \text{ Hz}$  or  $1730 \text{ Hz}$  or  $2883 \text{ Hz}$ . (Any multiples of  $576.7 \text{ Hz}$  are acceptable.)

- 8 a It is a three-dimensional wave that loses energy as it travels through the Earth's crust, with its intensity inversely proportional to the square of the distance travelled.

b  $I \propto \frac{1}{r^2}$

$\therefore Ir^2 = \text{constant}$

$\therefore I_1(r_1)^2 = I_2(r_2)^2$

$I_2 = \frac{I_1(r_1)^2}{(r_2)^2}$

$= \frac{(1.0 \times 10^6) \times (100)^2}{(500)^2}$

$= 4.0 \times 10^4 \text{ W m}^{-2}$

- 9 a For the first 30 s, the cyclist travels 150 m east at constant speed, then she accelerates for the next 10 s travelling a further distance of 150 m. She then travels at a higher constant speed for the next 10 s, travelling a further distance of 200 m.

b  $5 \text{ ms}^{-1}$  east

c  $25 \text{ ms}^{-1}$  east

d  $v_{\text{av}} = \frac{u+v}{2}$   
 $= \frac{5+25}{2}$

$= 15 \text{ ms}^{-1}$  east

e  $a = \frac{v-u}{t} = \frac{(25-5)}{10} = 2.0 \text{ ms}^{-2}$  east

f  $v_{\text{av}} = \frac{x}{t} = \frac{500}{50} = 10 \text{ ms}^{-1}$

10 a  $E_k = \frac{1}{2}mv^2$   
 $= \frac{1}{2} \times (2000) \times (40.0)^2$   
 $= 1.6 \text{ MJ}$

b  $\Sigma F = ma$   
 $= m \frac{(v-u)}{t}$   
 $= 2000 \frac{(40.0-0)}{5.0}$   
 $= 2000 \times 8.0$   
 $= 1.6 \times 10^4 \text{ N}$

c  $\Sigma F = 16000 \text{ N} = F - 400 \text{ N}$   
 $\therefore F = 16.4 \text{ kN}$

d  $W = Fx = (1.64 \times 10^4) \times (100) = 1.64 \times 10^6 \text{ J} = 164 \text{ MJ}$

**e**  $P = \frac{W}{t} = \frac{1.64}{5.0} = 328 \text{ kW}$

**f**  $W = 400 \times 100 = 40 \text{ kJ}$

**g** Efficiency =  $\frac{1.6}{1.64} \times 100 = 97.6\%$

**11 a**  $v^2 = u^2 + 2as$

$(8.0 \times 10^2)^2 = 0^2 + 2a \times 20$

$\therefore a = 1.6 \times 10^4 \text{ ms}^{-2}$

**b**  $F = ma = 550 \times 1.6 \times 10^4 = 8.8 \times 10^6 \text{ ms}^{-2}$

**c**  $p = mv = 550 \times 8.0 \times 10^2 = 4.4 \times 10^5 \text{ kg ms}^{-1}$

**d** Momentum of shell = momentum of gun

$(1.08 \times 10^5) \times v = 4.4 \times 10^5 \text{ kg ms}^{-1}$

$v = 4.1 \text{ ms}^{-1}$

**e**  $F = \frac{\Delta p}{\Delta t}$

$t = \frac{v - u}{a} = \frac{8.0 \times 10^2}{1.6 \times 10^4} = 0.05 \text{ s}$

$F = \frac{4.4 \times 10^5}{0.05} = 8.8 \times 10^6 \text{ N as before}$

**f** Work =  $F \times s$

$= 8.8 \times 10^6 \times 20$

$= 1.8 \times 10^8 \text{ J}$

**g**  $E_k \text{ of shell} = \frac{1}{2}mv^2 = 1.8 \times 10^8 \text{ J}$

This obviously represents an ideal situation; realistically there would be significant losses.

**12 a** 

**b**  $\lambda = \frac{2\ell}{5} = \frac{2 \times (1.50)}{5} = 0.600 \text{ m}$

$f = \frac{v}{\lambda} = \frac{(336)}{(0.600)} = 560 \text{ Hz}$

**c**  $\lambda = \frac{v}{f} = \frac{(336)}{(106)} = 3.17 \text{ m}$

$\ell = \frac{1\lambda}{2} = \frac{(3.17)}{2} = 1.59 \text{ m}$

The second didgeridoo is longer.

**13 a** The average velocity over that distance.

**b** A radar gun only gives the instant velocity at a given point in time. Timing over 100 m gives an average velocity over that distance.

**c** The skier accelerates down the slope due to a component of gravity,  $g \sin \theta$ , acting down the plane. The frictional force of the skis against the snow acts against this and retards the acceleration.

**d** Air resistance also acts against the motion of the skier down the slope, resulting in them reaching a terminal velocity when this resistance is equal to  $mg \sin \theta$ .

**e** The design of the skis acts to reduce the force on any given part of the snow under the skis by spreading the weight of the skier and equipment over a larger surface area: pressure =  $\frac{\text{force}}{\text{area}}$ .

**f**

| Design factor                 | How it reduces friction                                                                |
|-------------------------------|----------------------------------------------------------------------------------------|
| shape of the skis             | lower profile helps them to reduce wind resistance                                     |
| helmet design                 | directs the wind from the top of their head down their back while in the tuck position |
| shape of the boots            | low profile to lower wind resistance                                                   |
| skin-tight polyurethane suits | allows air to pass over the skier easily, reducing wind resistance                     |

**g** Acceleration is the rate of change of velocity. The faster they accelerate, the quicker they will reach terminal speed.

$$\text{h } v = \frac{254.958}{3.6} = 70.8217 \text{ ms}^{-1}$$

$$v_{av} = \frac{s}{\Delta t} \text{ and } v_{av} = \frac{v+u}{2}$$

$$\frac{s}{\Delta t} = \frac{v+u}{2}$$

$$\begin{aligned} \Delta t &= \frac{2s}{v+u} \\ &= \frac{2 \times 340}{70.8217 + 0} \\ &= 9.60 \text{ s} \end{aligned}$$

$$\begin{aligned} \text{i } a &= \frac{v-u}{t} \\ &= \frac{70.82 - 0}{9.60} \\ &= 7.38 \text{ ms}^{-2} \end{aligned}$$

$$\begin{aligned} \text{j } v_{av} &= \frac{s}{\Delta t} \\ \Delta t &= \frac{s}{v_{av}} \\ &= \frac{100}{70.82} \\ &= 1.41 \text{ s} \end{aligned}$$

$$\begin{aligned} \text{k } E_k &= \frac{1}{2}mv^2 \\ &= \frac{1}{2} \times (70.0) \times (70.8217)^2 \\ &= 1.76 \times 10^5 \text{ J} \end{aligned}$$

$$\begin{aligned} \text{l i } E_k &= E_p \times \frac{93}{100} \\ E_p &= (1.76 \times 10^5) \times \frac{100}{93} \\ &= 1.89 \times 10^5 \text{ J} \end{aligned}$$

$$\begin{aligned} \text{ii } E_p &= mgh \\ h &= \frac{E_p}{mg} \\ &= \frac{(1.89 \times 10^5)}{(70.0) \times (9.80)} \\ &= 275 \text{ m} \end{aligned}$$

$$\begin{aligned} \text{Altitude} &= 2720 - 275 \\ &= 2445 \text{ m} \end{aligned}$$

$$\begin{aligned} \text{m } W_d &= \Delta E \\ Fs &= E_p - E_k \\ F &= \frac{E_p - E_k}{s} \\ &= \frac{(1.89 \times 10^5) - (1.76 \times 10^5)}{(340)} \\ &= 38.2 \text{ N} \end{aligned}$$